

# Mikhail Kornev

Steklov Mathematical Institute of RAS

Nationality: Russian

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## RESEARCH AREAS

Geometry and Topology

## APPOINTMENTS

- Sep. 2022 – PhD student and research assistant at Steklov Mathematical Institute of RAS  
Department of Geometry and Topology
- Mar. 2026 – Research assistant at HSE University  
International Laboratory for Mirror Symmetry and Automorphic Forms
- Sep. 2021 – 2025 Research assistant at HSE University  
Laboratory of Algebraic Topology and Applications

## EDUCATION

- Sep. 2016 – May 2022 Specialist degree Diploma with honors in Mathematics (equivalent to M.S.)  
Lomonosov Moscow State University

## RESEARCH PAPERS

### Mathematics

- Jul. 2026 Buchstaber V.M., Kornev M.I.,  
*Algebraic  $n$ -valued groups, cubics, and Wendt matrices*  
Matematicheskii Sbornik, Volume 217, Number 8 (2026)  
<https://doi.org/10.4213/sm10344>
- Apr. 2026 Kornev M.I.,  
*Buchstaber, Ochanine, Krichever, and Witten genera*  
Journal of Geometry and Physics, Volume 226, Number 105855 (2026)  
<https://doi.org/10.1016/j.geomphys.2026.105855>
- Jan. 2026 Buchstaber V.M., Kornev M.I.  
 *$n$ -Valued Monoids on  $\mathbb{CP}^1$  and Discriminants*  
Russian Mathematical Surveys, Volume 81, Issue 1(487) (2026), 191–193  
<https://doi.org/10.4213/rm10280e>
- Jan. 2026 Kornev M.I.  
*On the Classification of  $n$ -Valued Monoids and Groups of Order 3*  
Mat. Zametki, Volume 119, Issue 2 (2026), 244–252  
<https://doi.org/10.4213/mzm14803>
- Jan. 2024 Kornev M.I.  
 *$n$ -Valued Coset Groups and Dynamics*

Mathematical Notes, Volume 116, Issue 1 (2024), 66–76  
<https://doi.org/10.4213/mzm14243>

Sep. 2023

Kornev M.I.  
*Homology Spheres*  
Transactions of the Moscow Mathematical Society, Volume 84 (2023), 197–239  
<https://doi.org/10.1090/mosc/352>

## Mathematics and Music

Sep. 2025

*The Technique of Musical Icosahedra*  
Music Academy, no. 3, 2025, pp. 64–83  
<https://doi.org/10.34690/479>

## PREPRINTS

Apr. 2026

Buchstaber V.M., Kornev M.I., Rubtsov V.N.  
*Two-Valued Groups, Chazy Equation, Dubrovin-Frobenius Structures, and QYBE*  
arXiv:2604.26641

Mar. 2026

Kornev M.I.,  
*Buchstaber, Ochanine, Krichever, and Witten genera*  
arXiv:2603.21118

Oct. 2025

Buchstaber V.M., Kornev M.I.  
*Algebraic  $n$ -Valued Monoids on  $\mathbb{CP}^1$ , Discriminants and Projective Duality*  
arXiv:2510.14010

Aug. 2025

Kornev M.I.  
*On the Classification of  $n$ -Valued Monoids and Groups of Order 3*  
arXiv:2508.04454

May 2025

Buchstaber V.M., Kornev M.I.  
 *$n$ -Valued Groups, Kronecker Sums, and Wendt's Matrices*  
arXiv:2505.04296

Jan 2024

Kornev M.I.  
 *$n$ -valued Coset Groups and Dynamics*  
arXiv:2401.12718

## CONFERENCE TALKS

May. 2026

*Buchstaber Genus and Formal Groups*  
Conference “Algebraic Topology, Group Actions, and Combinatorics,” Sirius, Sochi, Russia

Aug. 2025

*$n$ -Valued Groups, Kronecker Sums, and Wendt Matrices*  
joint work with Victor Buchstaber  
5th Conference of Mathematical Centers of Russia, Krasnoyarsk, Russia

May 2024

*Homology Spheres, Acyclic Groups and Kan-Thurston Theorem*  
Conference “From Analysis to Homotopy Theory”, University of Greifswald, Greifswald, Germany

Apr. 2023

*Dynamics and multivalued groups*  
Conference “SIMC Youth Race”, Steklov Mathematical Institute of RAS, Moscow, Russia

- Oct. 2022 *Inconcreteness of homotopy categories*  
International School “Toric topology, combinatorics and data analysis”, Euler  
International Mathematical Institute, Saint Petersburg, Russia
- Apr. 2022 *Acyclic groups and spaces*  
29th International Scientific Conference for Undergraduate and Graduate Students  
and Young Scientists “Lomonosov”, Moscow, Russia
- Nov. 2021 *Acyclic groups and Kan-Thurston theorem*, International School “Toric Topology and  
Combinatoric”, Sirius Mathematics Center, Sochi, Russia

## SEMINAR TALKS

- May 2026 *Buchstaber genus and formal groups*  
Seminar “Toric Topology, Combinatorics, and Homotopy Theory”  
jointly with the Laboratory of Topology and Complex Networks of the Faculty of  
Mathematics, HSE University,  
Moscow Center for Continuous Mathematical Education (MCCME), Moscow, Russia
- Dec. 2025 *Fermat hypersurfaces, projective duality, and algebraic  $n$ -valued monoids*  
Seminar of the Department of Geometry and Topology “Geometry, Topology and  
Mathematical Physics” (Novikov Seminar),  
Steklov Mathematical Institute of RAS, Moscow, Russia
- Dec. 2025 *Discrete and Algebraic  $n$ -Valued Groups*  
Seminar on Discrete Geometry and Geometry of Numbers,  
Moscow State University, Moscow, Russia
- Nov. 2025  *$n$ -Valued Algebraic Monoids, Cubics, and Discriminants*  
The Seminar of International Laboratory of Algebraic Topology and Its Applications,  
HSE University, Moscow, Russia
- May 2025  *$n$ -Valued Groups, Kronecker Sums, and Wendt Matrices*  
Joint work with Victor Buchstaber  
Seminar of the Department of Geometry and Topology “Geometry, Topology and  
Mathematical Physics”,  
Steklov Mathematical Institute of RAS (Novikov Seminar), Moscow, Russia
- Apr. 2023 *Acyclic groups and Kan-Thurston theorem*  
Seminar by I. G. Lysenok and A. L. Talambutsa “Seminar on geometric group theory”,  
Steklov Mathematical Institute of RAS, Moscow, Russia
- Dec. 2022 *Fundamental groups of homology spheres*  
Seminar “Geometry, Topology and their applications” at Sobolev Institute of  
Mathematics,  
Online

## GRANTS AND AWARDS

### Prizes, fellowships and grants

- 2021 – 2023 *The Russian Science Foundation Grant No. 23-11-00143*
- 2019 *Winner of the olympiad on Partial Differential Equations*  
Moscow State University
- 2018 *Winner of the olympiad on Ordinary Differential Equations*  
Moscow State University

## STUDY SEMINAR TALKS

### Texas Tech Quantum Homotopy Seminar

- Dec. 2024                      *An Introduction to Cartan Geometries*  
Mar. 2024                      *Euler-Lagrange dynamics via the  $\infty$ -jet bundle* (2 talks)

### Seminar On Homotopy Theory and Applications at Independent University of Moscow

- Dec. 2023                      *Principal  $\infty$ -bundles*  
Apr. – May 2023              *A mini-course on the  $\infty$ -category of motivic coarse spaces* (4 talks)  
Apr. 2023                      *Elements of unstable motivic homotopy theory*  
Feb. – Apr. 2023              *A mini-course on  $\infty$ -cosmoi* (4 talks)  
Dec. 2022                      *Applications of Reedy theory*  
Nov. 2022                      *Weighted (co-)limits, Gambino’s theorem, Reedy model structures*  
Nov. 2022                      *Homotopically final functors and examples of calculations with homotopy (co-)limits*  
Nov. 2022                      *Nerve—realization adjunction and bar construction*  
Apr. 2022                      *A series of 4 talks on sheaves of sets*  
Apr. 2022                      *Kan extensions. Additional topics*  
Oct. 2021                      *The categorical meaning of the Kan-Thurston theorem*  
Jul. – Aug. 2021              *Quillen’s theorem, a closed model structure on  $s\mathbf{Set}$*

### Study And Scientific Seminar “Geometry And Topology” at Moscow State University

- Sep. 2020                      *Andreev’s theorem on hyperbolic polyhedra*  
Nov. 2019                      *The intersection form of 4-manifolds*  
May 2019                      *Nilmanifolds and Massey product in bigraded cohomology rings*

## CONFERENCES, WORKSHOPS AND SCHOOLS

- May 2024                      *Conference “From Analysis to Homotopy Theory”*  
University of Greifswald, Greifswald, Germany  
Jul. 2023                      *Summer School “Algebra and Geometry”*  
HSE University, Steklov Mathematical Institute of RAS, Suzdal, Russia  
Jul. 2023                      *Summer school on non-commutative and derived geometry of matrix factorisations*  
Centre of Pure Mathematics, Moscow Institute of Physics and Technology,  
Dolgoprudny, Russia  
Jun. 2023                      *Conference on algebra, algebraic geometry, and number theory on the occasion of*  
*academician Igor Rostislavovich Shafarevich 100th birthday*  
Steklov Mathematical Institute of RAS, Moscow, Russia  
Apr. 2023                      *Conference “Geometry, topology and mathematical physics” on the occasion of the 85th*  
*anniversary of S. P. Novikov and the 80th anniversary of V. M. Buchstaber*  
Steklov Mathematical Institute of RAS, Moscow, Russia  
Mar. 2023                      *Conference “SIMC Youth Race”*  
Steklov Mathematical Institute of RAS, Moscow

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|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Oct. 2022 | <i>International School “Toric topology, combinatorics and data analysis”</i><br>Euler International Mathematical Institute, Saint Petersburg, Russia                         |
| Jul. 2022 | <i>Summer School “Algebra and Geometry”</i><br>HSE University, Steklov Mathematical Institute of RAS, Suzdal, Russia                                                          |
| Apr. 2022 | <i>29th International Scientific Conference for Undergraduate and Graduate Students and Young Scientists “Lomonosov”</i><br>Lomonosov Moscow State University, Moscow, Russia |
| Nov. 2021 | <i>International School “Toric Topology and Combinatorics”</i><br>HSE University, Sirius Mathematics Center, Sochi, Russia                                                    |
| Nov. 2020 | <i>Topology and Geometry of Group Actions</i><br>HSE University, online                                                                                                       |
| Nov. 2019 | 7th Summer School On Geometrical Methods Of Mathematical Physiques<br>Landau Institute for Theoretical Physics, Lomonosov Moscow State University, Chernogolovka, Russia      |
| Nov. 2019 | <i>Geometry, Topology And Mathematical Physics</i><br>Landau Institute for Theoretical Physics, Lomonosov Moscow State University, Chernogolovka, Russia                      |
| Nov. 2018 | <i>Geometry, Topology And Mathematical Physics</i><br>Landau Institute for Theoretical Physics, Lomonosov Moscow State University, Chernogolovka, Russia                      |
| Jul. 2018 | <i>Summer school “Modern Mathematics”</i><br>HSE University, Lomonosov Moscow State University, Independent University of Moscow, Dubna, Russia                               |
| Jul. 2017 | <i>Summer school “Modern Mathematics”</i><br>HSE University, Lomonosov Moscow State University, Independent University of Moscow, Dubna, Russia                               |

## TEACHING

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| May 2022 – 2024  | <i>Teaching various courses (e. g., Algebraic Geometry, Algebraic Topology, Abstract Algebra, Differential Geometry, Linear Algebra, Probability Theory, Discrete Mathematics, Calculus, etc.) on Profi</i><br><a href="https://profi.ru/profile/KornevMI3/">https://profi.ru/profile/KornevMI3/</a> |
| Sep. 2016 – 2022 | <i>Little Mech-Math courses</i><br>An olympiad Math class at Lomonosov Moscow State University<br><a href="http://mmmf.msu.ru/about/person.html">http://mmmf.msu.ru/about/person.html</a>                                                                                                            |

## SERVICE

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| June 2021 – 2024 | <i>Co-organization of the study seminar “Homotopy Theory and Applications” at Independent University of Moscow</i><br><a href="https://researchseminars.org/seminar/HomotopyTheoryAndApps">https://researchseminars.org/seminar/HomotopyTheoryAndApps</a> |
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## REFERENCES

*Victor Buchstaber,*

Steklov Mathematical Institute of RAS  
buchstab@mi-ras.ru

## **LANGUAGES**

*Russian* (native)

*English* (fluent)

*French* (intermediate)

*Latin* (basic)